

SE LAB1

PROBLEM STATEMENT 22: Smart Cities, Transport & Logistics

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1. Functional and Non-Functional requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall record vehicle entry time via license plate scan and compute total billing amount upon exit based on tiered hourly duration.	High	Pass: Accurate fee calc based on elapsed duration; Fail: wrong tier applied	Core billing functionality
FR-002	Functional	The system shall allow a vehicle owner to reserve a specific parking bay for a chosen date and time window before arrival.	High	Pass: Reservation is confirmed and bay is marked "reserved" in the system within 5 seconds of submission; Fail: bay remains shown as available	Enables advance planning and reduces search time
FR-003	Functional	The system shall validate a scanned license plate against an active reservation before opening the entry gate.	High	Pass: Gate opens only when plate matches an active, unexpired reservation; Fail: gate opens for unmatched/expired plate	Prevents unauthorized bay occupation
FR-004	Functional	The system shall allow the vehicle owner to pay the calculated parking fee via a digital payment method (card/UPI/wallet) at exit.	High	Pass: Payment confirmation received and gate opens within 10 seconds of successful transaction; Fail: gate opens without payment confirmation, or gate doesn't open after payment confirmation.	Enables cashless, frictionless exit
FR-005	Functional	The system shall automatically flag a vehicle to the parking warden if it remains parked beyond its reserved/paid duration by more than a 15-minute grace period.	Medium	Pass: Alert appears on warden's dashboard within 1 minute of grace period expiry; Fail: no alert generated for an overstaying vehicle	Enforces fair bay turnover and prevents revenue loss

Table 1: Functional Requirements.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The gate camera license plate lookup API shall verify parking reservation validity in under 300 ms.	High	Pass: Benchmarking confirms target latency and security standards under simulated peak load	Keeps entry gate throughput fast, avoids queues
NFR-002	Security	All payment transaction data transmitted between the app, payment gateway, and backend shall be encrypted using TLS 1.2+ in transit and AES-256 at rest.	High	Pass: Security audit/penetration test confirms no plaintext transmission or storage of payment data; Fail: any unencrypted payment data found	Protects sensitive financial data and ensures regulatory compliance

Table 2: Non - Functional Requirements.

- 2. Actors: Vehicle Owner, Parking Warden, Payment Gateway
- 3. Use cases and UML Use- case diagram:

ID	Use Case	Primary Actor
UC-01	Reserve Parking Bay	Vehicle Owner
UC-02	Scan License Plate at Entry	Vehicle Owner (triggers gate camera)
UC-03	Validate Reservation	(internal)
UC-04	Calculate Parking Fee	(internal)
UC-05	Exit & Pay Parking Fee	Vehicle Owner, Payment Gateway
UC-06	Apply Discount Code	Vehicle Owner
UC-07	Flag & Resolve Unauthorized Overstay	Parking Warden

Table 3: Use cases.

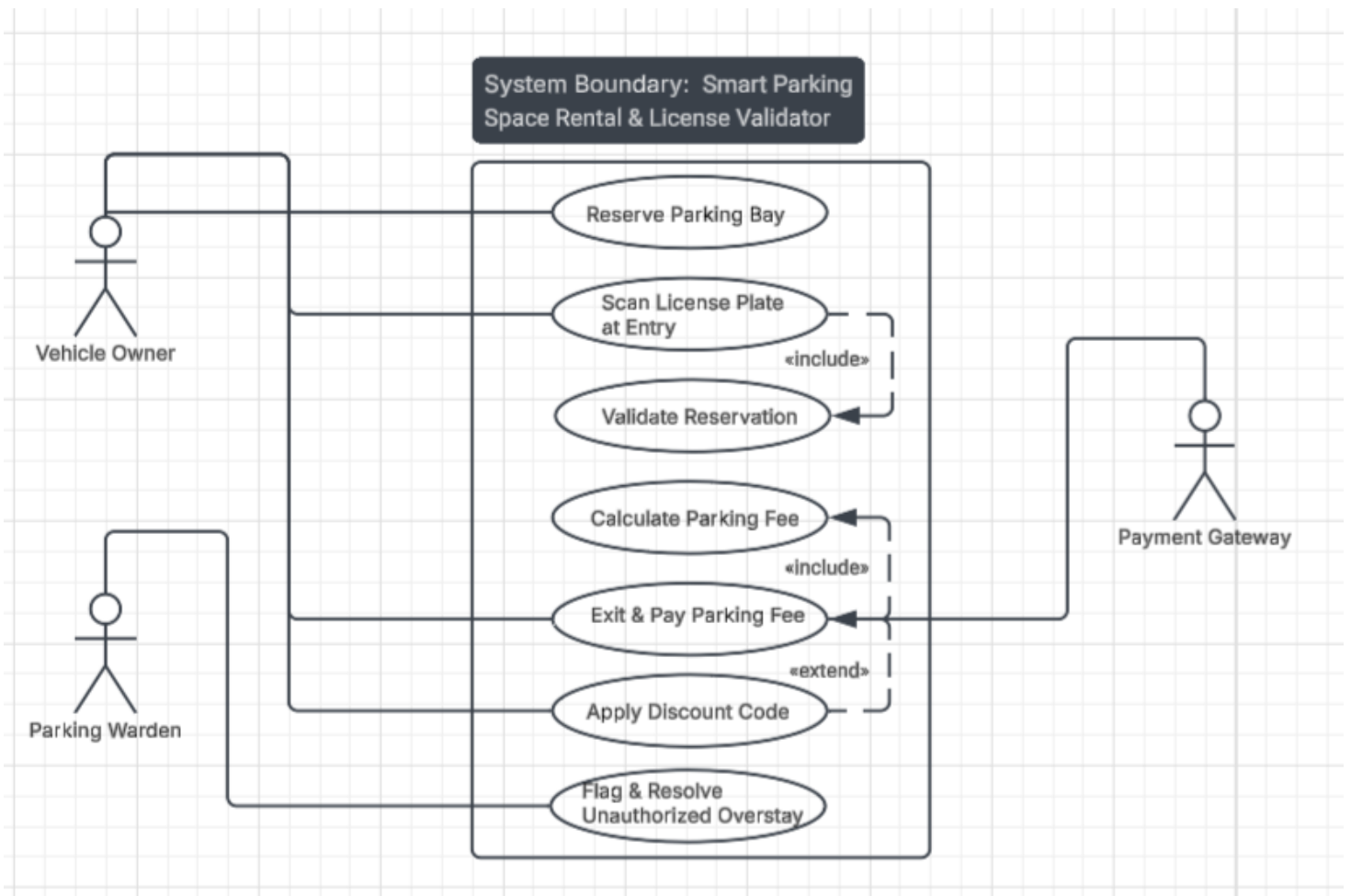


fig 1: UML Use case Diagram.

4. Use-Case Flow

Use Case: Exit & Pay Parking Fee

Preconditions:

- Vehicle has a logged entry record with timestamp (from FR-001).
- Vehicle owner is physically at the exit gate.

Postconditions:

- Parking fee is calculated and charged.
- Exit time is logged.
- Exit gate opens and bay is marked "available."

Main Success Scenario:

1. Vehicle owner drives up to the exit gate; the gate camera scans the license plate.
2. System retrieves the vehicle's entry timestamp and matches it to the logged record.
3. System calculates elapsed duration and computes the total fee using tiered hourly rates (includes UC-04: Calculate Parking Fee).

4. System displays the fee amount on the exit kiosk/app.
5. Vehicle owner selects a payment method (card/UPI/wallet).
6. System connects to the Payment Gateway and requests authorization.
7. Payment Gateway authorizes the transaction.
8. System logs the exit time and marks the bay as available.
9. A payment confirmation/receipt is displayed and/or emailed to the owner.
10. Exit gate opens; use case ends successfully.

Alternate Flow - 7a. Payment Declined:

- 7a1. System displays a "Payment Failed" message on the kiosk/app.
- 7a2. Vehicle owner is prompted to select another payment method.
- 7a3. If payment fails again after 2 attempts, the system alerts the parking warden and holds the gate closed until manual resolution.